

William Reed

📍 2A Carswell St, Waco, TX 76705 📞 +1-512-737-3009 ✉ wreed0813@gmail.com 🌐 <https://www.linkedin.com/in/william-reed-342686385>

Summary

A highly capable Software Engineer with significant experience in transitioning to a Golang Engineer position, renowned for developing scalable cloud-native solutions with an emphasis on backend architecture. Demonstrates proficiency in building microservices and real-time data processing systems with AWS and Redis, underscoring a commitment to backend resilience and efficiency. Eager to apply deep industry knowledge and drive impactful results as a Golang Engineer, enhancing system performance and robustness.

Technical Skills

- **Skills:** React 18, Next.js, Redux, TypeScript, Tailwind CSS, Styled Components, Framer Motion, GSAP, Node.js, Express.js, REST APIs, GraphQL, OAuth2, JWT, Azure AD, Okta, Google Auth, FHIR, HL7, React Hook Form, Accessibility (WCAG 2.1 AA), SSR, Component Libraries, Docker, EKS, Terraform, Agile, CI/CD, Blue/Green Deployment, React, React Native, TypeScript, GraphQL, REST APIs, FHIR, HL7, React Hook Form, i18n, Tailwind CSS, Custom UI Libraries, PWA, Service Workers, Firebase Cloud Messaging, GitHub Actions, AKS, Docker, CI/CD Pipelines, Mobile-first Design, Real-time Messaging, React, Redux, Node.js, MySQL, Redis, REST APIs, OpenAPI/Swagger, Tailwind CSS, Performance Optimization, Admin Portals, Third-party Integrations, GraphQL (implied from structure), Technical Documentation, Agile Development, API Caching, Node.js, JWT, Angular, D3.js, Google Charts, REST APIs, Three.js, Next.js, SSR, Lazy Loading, Code Splitting, Mocha, Chai, Postman, Modular Services, Multi-tenant SaaS, Dashboard UI, Real-time Data Visualization
- **Frontend Development:** React 18, Next.js (SSR & SEO), React Native, Redux, TypeScript, Tailwind CSS, Styled Components, Framer Motion, GSAP, Angular (up to 17), Bootstrap, Material UI, React Hook Form, i18n, PWA, Accessibility (WCAG 2.1 AA)
- **UI/UX Engineering:** Design Systems, Component Libraries, Figma Integration, Responsive Design, Microinteractions, Mobile-first Layouts, Modular Component Design, Form Builders, Admin Dashboards
- **Backend & API Development:** Node.js, Express.js, .NET Core (API Layer), GraphQL (Apollo, HotChocolate), REST APIs, Prisma ORM, OAuth2, JWT, RBAC, SignalR, gRPC
- **Healthcare & Data Standards:** FHIR, HL7, Real-Time EHR Dashboards, Secure Messaging, HIPAA-compliant Development
- **Cloud Platforms & Infrastructure:** AWS (Lambda, ECS, S3, EventBridge), Azure (Functions, Event Grid, App Service, SQL DB), Kubernetes (AKS, EKS), Docker, Terraform, Firebase Cloud Messaging, Blue/Green Deployments
- **Testing & Quality Assurance:** Jest, Cypress, Playwright, Selenium, xUnit, NUnit, Mocha, Chai, Postman, TDD, BDD, CI/CD Validation Pipelines
- **Databases & Caching:** MySQL, PostgreSQL, SQL Server, Cosmos DB, MongoDB, Redis, Firebase Firestore
- **DevOps & CI/CD:** GitHub Actions, Azure DevOps, Helm, Dockerfile/Containerization, Git (GitHub, Bitbucket), API Versioning & Documentation (OpenAPI)
- **Other Tools & Practices:** Agile (Scrum, Kanban), Mentorship & Code Reviews, Real-Time Dashboards, WebSockets, Component Refactoring, Cross-Platform Development

Experience

Coherent Solutions

Jul 2023 - Present

Senior Software Engineer

Minneapolis, MN

- Architected scalable microservices capable of processing simulated sensor payloads at ~100K events/minute using BullMQ, Redis, and AWS EventBridge, mimicking ADAS workloads.
- Built real-time geospatial dashboards using coordinate-based datasets and WebSocket feeds, supporting map-based filtering and region detection for autonomous logistics workflows.
- Designed a feature flag and A/B testing pipeline powered by AWS Lambda and LaunchDarkly to test ML-assisted UI workflows and contextual recommendations across 10K+ daily users.
- Developed data ingestion and transformation services that processed hundreds of GBs per day, including telemetry, event logs, and geospatial activity data across multiple tenants.
- Created a reusable canvas module using Konva.js for rendering annotations over large map and sensor data tiles in web-based debugging tools. Delivered modular, scalable backend services using NestJS, improving code maintainability and enabling faster onboarding for new developers across services.
- Architected an event-driven job queue using BullMQ and Redis, handling 10,000+ background tasks daily with retry logic, priority handling, and dead-letter queues.
- Integrated AWS EventBridge to decouple services and orchestrate async workflows across auth, logging, and user engagement flows, improving system resilience.
- Identified and resolved backend bottlenecks using Node.js profiling tools (Clinic.js, Chrome DevTools), reducing cold start times by 40% and API latency by 25%.
- Tuned critical API paths with PostgreSQL and MongoDB optimizations, leveraging connection pooling and indexed queries to boost backend throughput during peak traffic.

- Spearheaded peer code review standards for backend services, reducing post-release defects by 20% and mentoring junior developers on clean code and API architecture.
- Defined best practices for NestJS service structure, middleware pipelines, and API versioning, accelerating squad development velocity by 15%.
- Led full-stack development using Node.js (Express), React 18, and Next.js, delivering core product features across Agile bi-weekly sprints.
- Built SSR-optimized pages in Next.js with dynamic routing, significantly improving SEO and page load performance.
- Developed AI-powered personalization features in React, surfacing contextual recommendations and smart form suggestions, boosting user engagement.
- Created reusable React + Redux admin interfaces, reducing internal support ticket resolution time by 30%.
- Built and maintained a custom React-based SSO onboarding flow with OAuth2 integrations (Azure AD, Okta, Google), simplifying enterprise access.
- Developed internal design systems with React and Styled Components, driving theme consistency across microfrontends.
- Delivered polished, accessible UI experiences using React, Tailwind CSS, and WCAG 2.1 AA-compliant practices.
- Enhanced UX with rich animations using Framer Motion and GSAP, improving perceived performance and microinteractions.
- Implemented React Hook Form with custom validation logic across forms, reducing user input errors and abandonment.
- Collaborated with stakeholders to launch an A/B testing framework with feature flags, enabling data-informed UX decisions.
- Integrated REST and GraphQL APIs into real-time UIs, including healthcare data from FHIR and HL7 sources.
- Integrated AWS SQS queues with Node.js microservices for asynchronous task processing, incorporating dead-letter queues and retry logic for resiliency.
- Leveraged AWS SNS to decouple service-to-service communications, enabling real-time user notifications and audit logging.
- Monitored Lambda functions, API Gateway routes, and ECS tasks using AWS CloudWatch dashboards and alarms with custom metrics and filters.
- Automated multi-environment infrastructure deployments using Terraform and AWS CloudFormation, improving consistency across dev, staging, and production.
- Implemented secure secret management with AWS KMS, integrating encrypted env variables in containerized and Lambda-based workloads.
- Utilized Linux CLI tools (journalctl, top, netstat, curl) for diagnosing containerized microservices on AWS ECS and Lambda, improving incident response time by 40%.
- Designed Jenkins pipelines for Node.js and Docker-based services, incorporating automated tests, container builds, and deployment workflows to ECS and Lambda environments.
- Configured ALB and Route 53 for traffic routing and load balancing in blue/green deployments, optimizing release cycles and rollback strategies.

iConnect Consulting

Jan 2022 - Jun 2023

Senior Software Engineer

Orlando, FL

- Built scalable React (TypeScript) modules for appointment scheduling, real-time messaging, and EHR data views, improving patient and provider satisfaction.
- Designed and developed a mobile-first React Native check-in and booking experience, streamlining iOS and Android patient workflows.
- Created unified data layers with GraphQL and REST and integrated them into React UIs, enabling seamless interaction with EHR and insurance systems.
- Created GraphQL APIs for syncing insurance and patient data from FHIR and HL7 feeds.
- Introduced custom form builders using React Hook Form and i18n support for multilingual accessibility.
- Standardized component design through custom UI libraries using React and Tailwind, enhancing maintainability and visual coherence.
- Built a prototype healthcare assistant app using Firebase Firestore for document-based data modeling and Firebase Auth for secure login workflows.
- Implemented RBAC rules and Firestore queries to serve personalized dashboards and real-time health notifications.
- Optimized user input flows using React Hook Form, improving validation logic and decreasing form abandonment rates.
- Collaborated with ML engineers to expose model inference results in real-time React UIs, building admin tooling for labeling and adjusting prediction outputs.
- Successfully integrated telemetry-like feeds into real-time dashboards, enabling clinicians and support staff to enhance patient monitoring and improve accuracy by simulating autonomous driver system conditions.

Levvel

Jan 2019 - Nov 2021

Full Stack Developer

Charlotte, NC

- Built end-to-end features for logistics and e-commerce clients using Node.js, React, and MySQL, delivering consistent sprint-based improvements.
- Designed and deployed scalable REST APIs and third-party integration layers, increasing external API usage by 60%.
- Developed dynamic admin portals using React + Redux, streamlining support workflows and reducing ticket resolution time by 30%.
- Tuned backend performance with Redis caching and optimized queries, cutting API response times by over 60%.
- Created and maintained OpenAPI specs and internal documentation, which improved developer onboarding and reduced QA cycles.
- Designed customer insight dashboards using Google Charts and D3.js to visualize user behavior, admin KPIs, and service utilization trends across regions.

- Enabled interactive filtering, real-time data updates, and exportable analytics views to empower business and product teams in feature planning.
- Contributed to CI/CD automation using Jenkins and BitBucket Pipelines, enabling container image builds, artifact promotion, and ECS task updates.
- Assisted in provisioning AWS infrastructure using Terraform for service-specific environments and ECS cluster scaling policies.
- Created shell scripts to automate dev environment setup and log aggregation tasks, streamlining onboarding and reducing setup time by 50%.
- Partnered with DevOps to migrate services from Bitbucket Pipelines to GitLab CI, implementing YAML-based build pipelines with environment-specific deploy triggers and rollback strategies.
- Developed and executed a pilot using AWS Neptune to uncover graph-based insights in logistics data, significantly enhancing decision-making efficiency and optimizing routing strategies.
- Enhanced route optimization APIs and vector-based overlays to significantly improve path visualization accuracy and streamline delivery workflows, leveraging ADAS guidance informed by data patterns.
- Built dynamic heatmaps and vector overlays using D3.js and Google Maps APIs to represent traffic congestion and fleet positioning in real time.

Pariveda Solutions

Aug 2014 - Nov 2018

Full Stack Developer

Dallas, TX

- Modernized legacy systems using Node.js, replacing outdated batch scripts and APIs with faster, modular services.
- Built authentication and session layers using JWT, enabling secure access and multi-tenant SaaS workflows.
- Created responsive dashboards using Angular and connected them to REST APIs for real-time logistics and fleet tracking.
- Configured email infrastructure for transactional notifications using SendGrid and implemented SPF/DKIM/DMARC DNS records to improve deliverability and lower spam risk.
- Built Node.js services for asynchronous email delivery using BullMQ and Redis queues, with retry logic and monitoring dashboards for bounce/spam analysis.
- Designed interactive data visualizations using D3.js and Google Charts, used by operational leads for daily decision-making.
- Developed interactive 3D environment previews and object overlays using Three.js and WebGL, supporting spatial planning interfaces used in warehouse robotics systems.
- Created modular canvas tools using Konva.js for real-time annotation over camera feeds and sensor-derived spatial data.
- Developed 3D product preview modules using Three.js for warehouse planning interfaces.
- Built SSR storefront pages in Next.js with lazy loading, code splitting, and performance tuning.
- Implemented testing suites using Mocha, Chai, and Postman, which decreased post-release bugs by 20%.
- Deployed services within custom AWS VPCs using NAT Gateways and security groups to control internal communication and egress traffic.
- Configured DNS routing for customer subdomains using Route 53, supporting multi-tenant SaaS onboarding flows.
- Managed transactional email systems using SendGrid with SPF/DKIM/DMARC DNS records, improving deliverability and compliance.

Education

Dallas Baptist University

2010 - 2014

Bachelor of Science, Computer Science